Predicting Problematic Internet Usage

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Introduction

This project stems from the Kaggle Competition, "Problematic Internet Use" by the Child Mind Institute. The goal is to predict the level of each participant's Severity Impairment Index (`SII`) score, which is a measure of problematic internet use by children and adolescents. Most predictor variables have to do with the participants' physical activity and online habits. We also wanted to see if we could predict the Sleep Disturbance Scale (`SDS`) score, which measures quality of sleep and is used in diagnosing certain conditions. We built multiple machine learning models with these goals in mind.

Handling NA Values

The data set that we were given for the competition had a lot of NA values. We tried a couple different methods for dealing with the NA values in our data set. For our linear regression approach, there were a few steps in our process. First, we removed the variables that had more than 50 percent NA values. There were only 10 variables that fit this description, but they all had a large proportion of NA values and we felt it would be best if we excluded those variables. Then, we removed all observations that had more than 50 percent NA values. There were 1097 variables removed this way, with a surprising large number of those observations having more than 90 percent NA values. Finally, we used imputation to fill in the remaining NA values with the "mice" package. Even after using imputation, there were still some NA values. We tried using a couple different methods to see if that was the issue, but still ended up getting NAs. Luckily, we noticed that all of the NA values were in the same columns and rows. For example, any observations that had an NA value in Fat Free Mass also had NA values in Fat Free Mass Index and Fat Mass Index. So we had a choice of either removing the observations with the NAs or the variables with the NAs, and we chose to remove the observations rather than removing more variables. After cleaning up the data with this method, we had 61 variables, 1584 observations in the training set, and 360 observations in the test set.

Variable	Total_NAs	NA_Proportion
Physical.Waist_Circumference	3062	0.773
Fitness_Endurance.Max_Stage	3217	0.812
Fitness_Endurance.Time_Mins	3220	0.813
Fitness_Endurance.Time_Sec	3220	0.813
FGC.FGC_GSND	2886	0.729
FGC.FGC_GSND_Zone	2898	0.732
FGC.FGC_GSD	2886	0.729
FGC.FGC_GSD_Zone	2897	0.732
PAQ_A.PAQ_A_Total	3485	0.880
PAQ_C.PAQ_C_Total	2239	0.565

ID	Total_NAs	NA_Proportion
0016bb22	58	0.951
0083e397	58	0.951

ID	Total_NAs	NA_Proportion
0087dd65	58	0.951
0194c251	58	0.951
023e1b6a	58	0.951
040d103f	58	0.951
04ccb838	58	0.951
05dc60ef	58	0.951
0677d22d	58	0.951
0882b065	58	0.951

For the other methods, we tried removing observations with a missing value for `SII` score and imputing the rest of the NA values. We are interested to see which of these methods gives us better results.

The Models

Linear Regression

Since the `SII` score for the participants is a categorical variable, we couldn't use linear regression to predict `SII` score. While `SDS` score is confined to only integers, it's not a categorical variable, so we can use linear regression to make predictions on that variable.

The first linear regression method we used was Lasso regression. Lasso regression suggested that we only use 28 variables to predict `SDS` score. It was interesting to see that only 2 of the 17 Bioelectric Impedance Analysis (BIA) variables were used. BIA measures things like body mass index, fat free mass, extracellular water, lean dry mass and other things of that nature. The only two BIA variables that were used in this model were Body Mass Index and Activity Level. BMI had a very small positive estimated coefficient and Activity Level had a relatively high negative estimated coefficient, which makes sense since it's generally better to be more active and have a lower BMI. After using cross validation to find the best value for λ, our best Lasso model ended up with a test root mean squared error (RMSE) of 10.73954.

The other linear regression method we used was Ridge regression. Ridge is very similar to Lasso, but does have a slightly different variable selection process that can result in a different "best model" than Lasso. In this case, the exact same predictors were used in both models, just with different coefficients associated with those predictors. The estimated coefficients for BMI and Activity level were both very similar to the results we got when using Lasso regression. After using cross validation again to find the best value for λ, our best Ridge model ended up with a Test RMSE of 10.6742.

```
## RMSE for LASSO Regression: 10.7397
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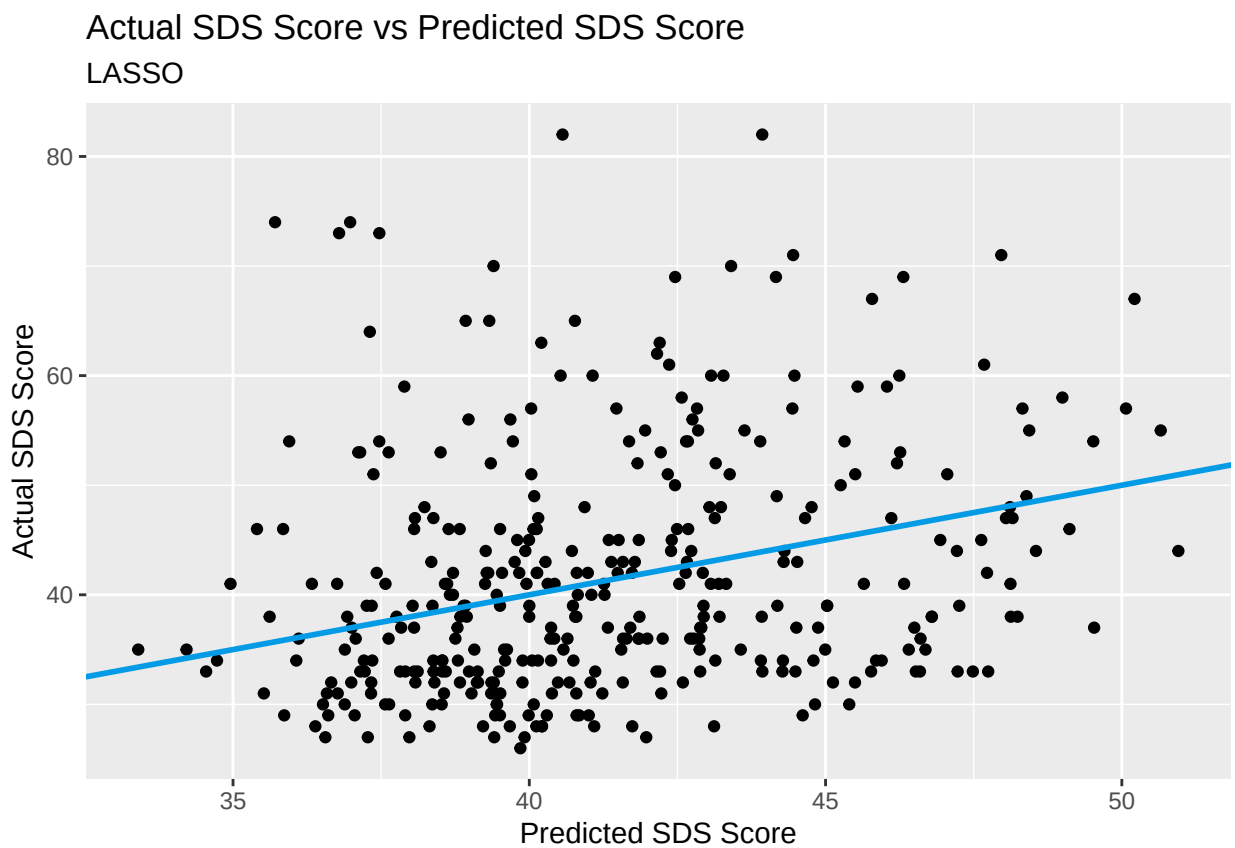


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## RMSE for Ridge Regression: 10.6802
```
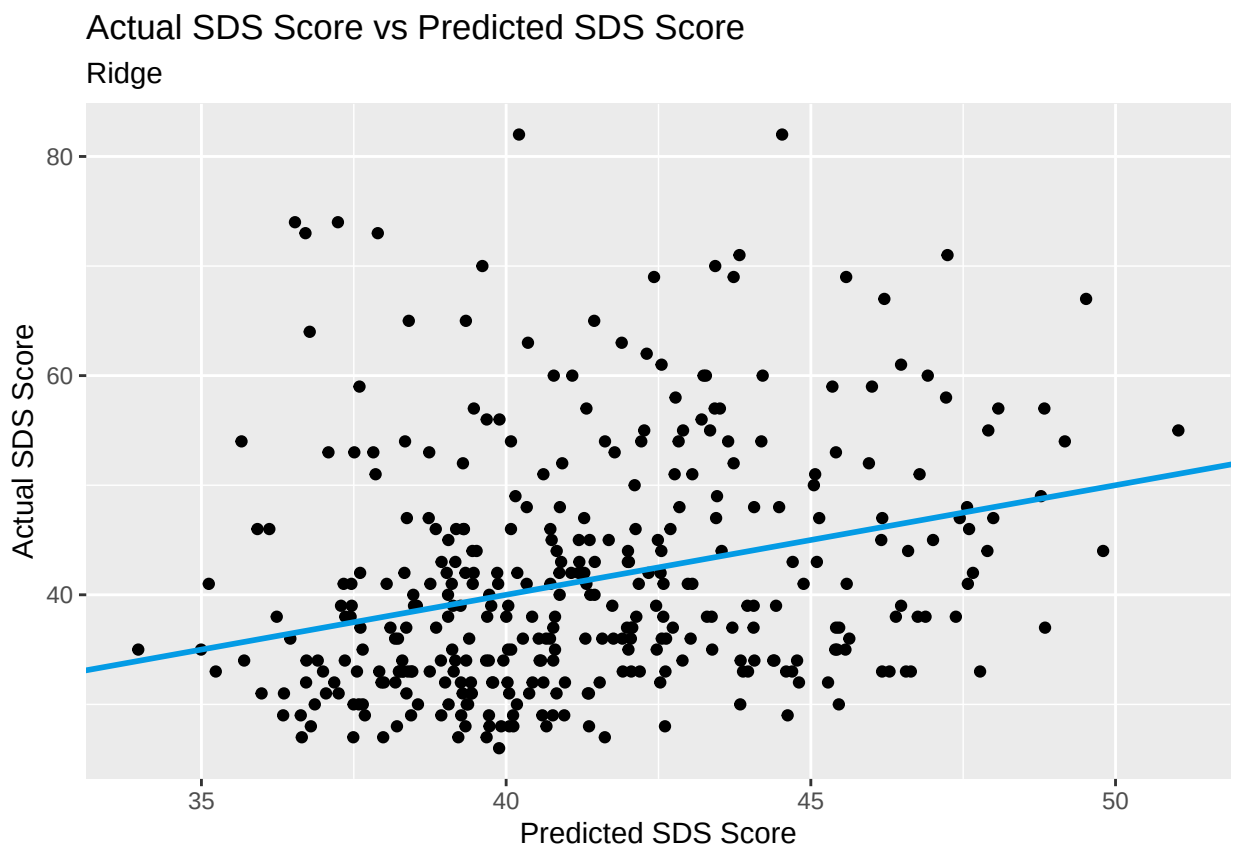



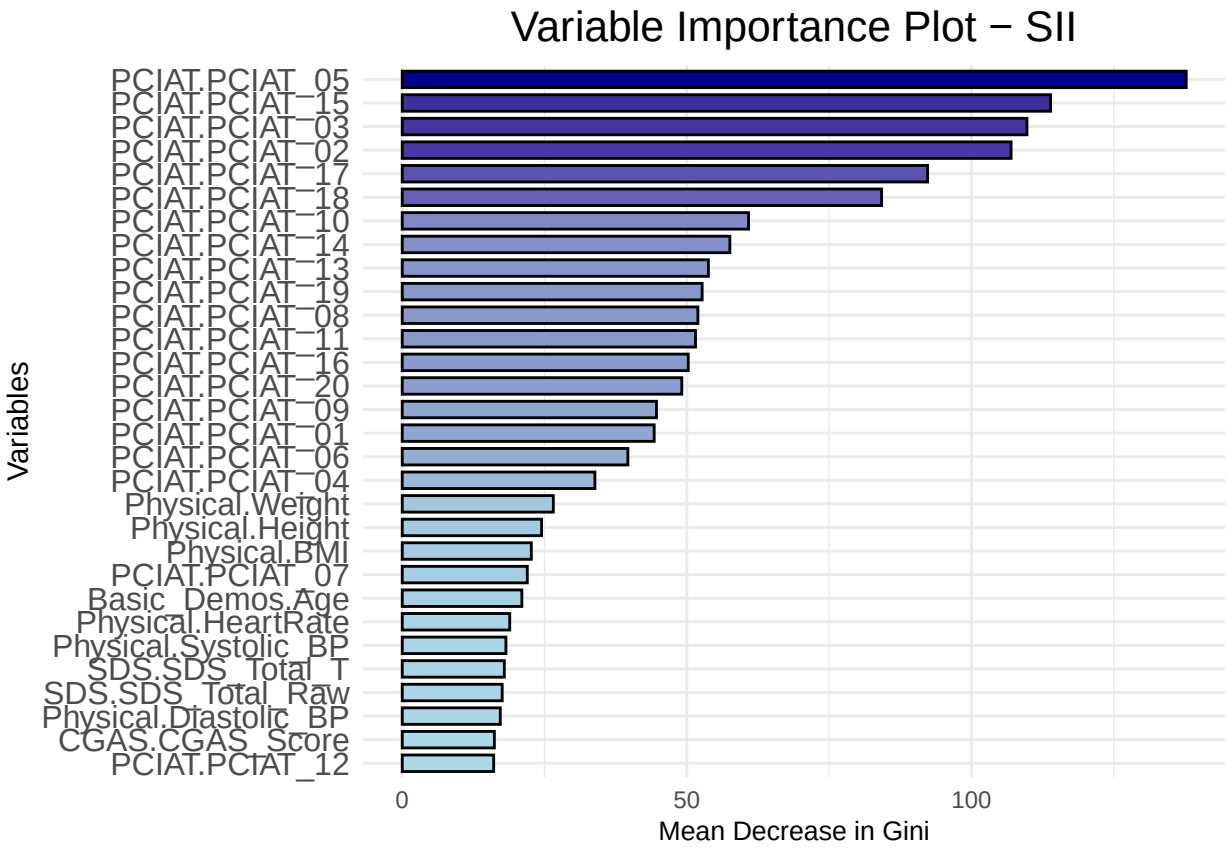
Random Forest

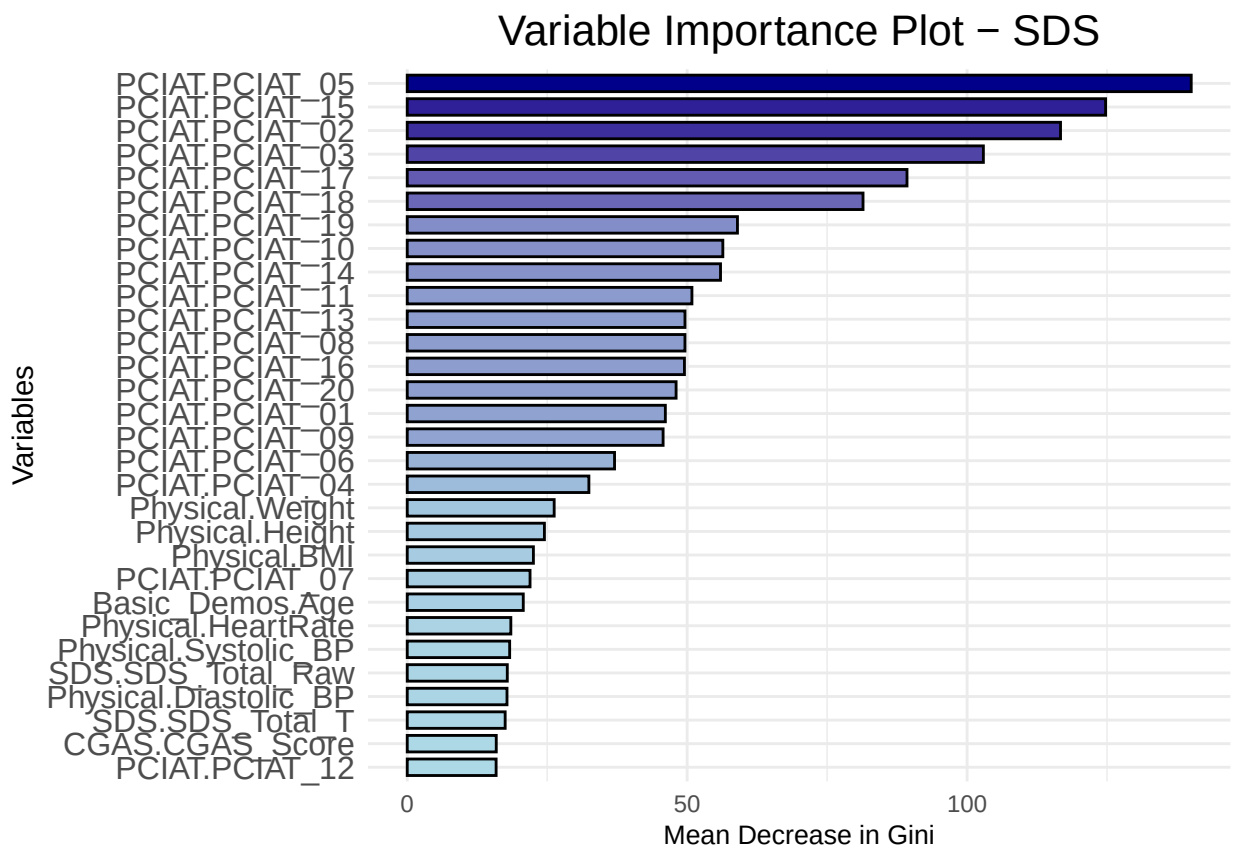
Random Forest is a classification and regression method used to predict both continuous and categorical variables. In this project, we built two Random Forest regression models to predict the Severity Impairment Index (SII) and Sleep Disturbance Scale (SDS). Data preprocessing involved removing variables with over 40% missing data, reducing the dataset by 36 variables and 1,225 observations. The PCIAT total score was excluded due to redundancy, but individual PCIAT variables were retained.The models showed that higher PCIAT scores are strongly associated with both SII and SDS, indicating that participants with stronger agreement on PCIAT statements tend to have higher severity impairment and sleep disturbance levels. Importance plots revealed PCIAT variables as the strongest predictors, with physical measures such as BMI, weight, height as secondary contributors. These findings highlight the significant role of PCIAT variables in predicting impairment and sleep disturbance issues, supported by physical measures for additional context.

Table 3: Variable Importance:

	MeanDecreaseGini
Basic_Demos.Enroll_Season	6.330932
Basic_Demos.Age	21.012790
Basic_Demos.Sex	2.914283
CGAS.Season	7.960081
CGAS.CGAS_Score	16.184930
Physical.Season	7.304582
Physical.BMI	22.678469
Physical.Height	24.476188
Physical.Weight	26.537186

	MeanDecreaseGini
Physical.Diastolic_BP	17.219894
Physical.HeartRate	18.882796
Physical.Systolic_BP	18.217416
Fitness_Endurance.Season	5.737302
FGC.Season	7.636837
BIA.Season	7.706932
PAQ_A.Season	4.194808
PAQ_C.Season	6.933553
PCIAT.Season	5.761960
PCIAT.PCIAT_01	44.260501
PCIAT.PCIAT_02	106.948409
PCIAT.PCIAT_03	109.741972
PCIAT.PCIAT_04	33.846048
PCIAT.PCIAT_05	137.714696
PCIAT.PCIAT_06	39.634333
PCIAT.PCIAT_07	21.967590
PCIAT.PCIAT_08	51.935604
PCIAT.PCIAT_09	44.654834
PCIAT.PCIAT_10	60.834523
PCIAT.PCIAT_11	51.514253
PCIAT.PCIAT_12	16.060530
PCIAT.PCIAT_13	53.779679
PCIAT.PCIAT_14	57.554250
PCIAT.PCIAT_15	113.896207
PCIAT.PCIAT_16	50.240008
PCIAT.PCIAT_17	92.289406
PCIAT.PCIAT_18	84.206712
PCIAT.PCIAT_19	52.680135
PCIAT.PCIAT_20	49.112891
SDS.Season	7.202872
SDS.SDS_Total_Raw	17.555024
SDS.SDS_Total_T	17.931543
PreInt_EduHx.Season	7.064890
PreInt_EduHx.computerinternet_hoursday	9.429413

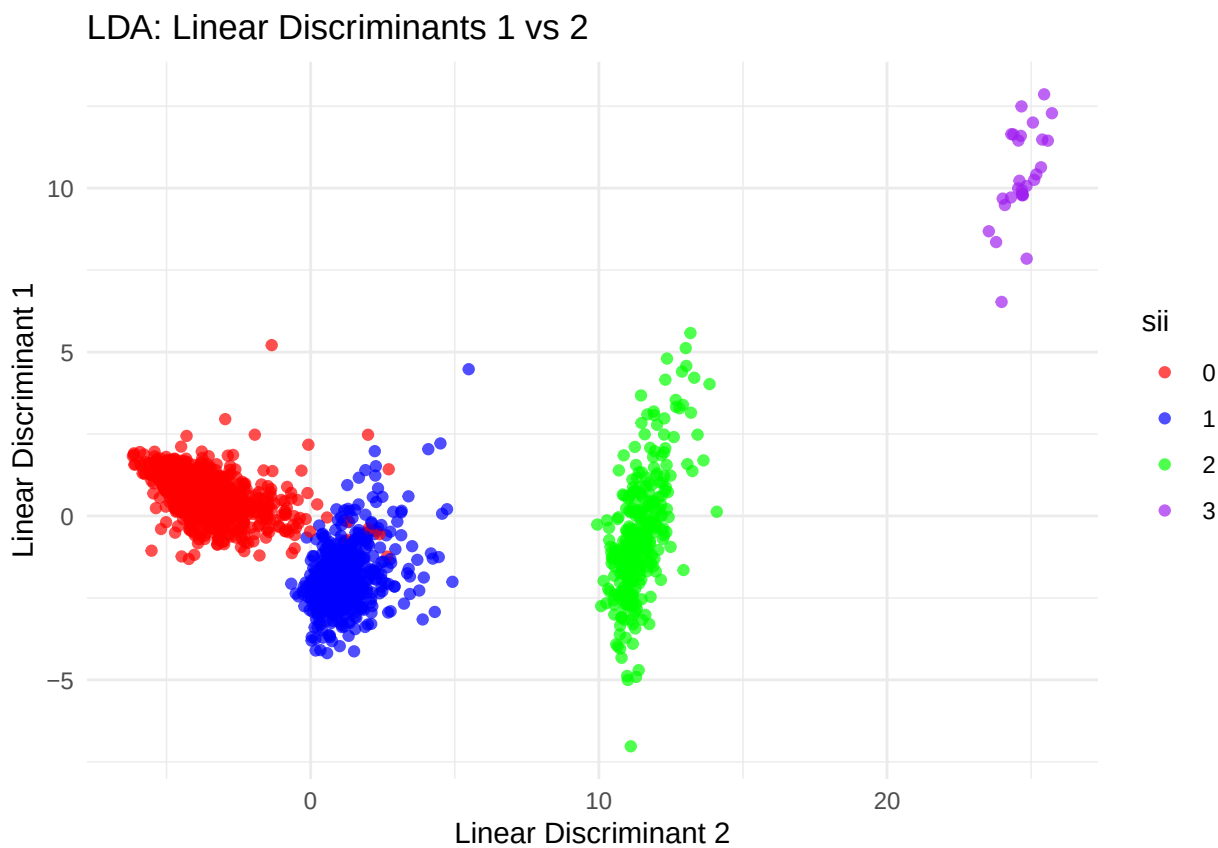

Variable Importance Plot − SII
PCIAT.PCIAT_05
PCIAT.PCIAT_15
PCIAT.PCIAT_03
PCIAT.PCIAT_02
PCIAT.PCIAT_17
PCIAT.PCIAT_18
PCIAT.PCIAT_10
PCIAT.PCIAT_14
PCIAT.PCIAT_13
PCIAT.PCIAT_19
PCIAT.PCIAT_08
PCIAT.PCIAT_11
PCIAT.PCIAT_16
PCIAT.PCIAT_20
PCIAT.PCIAT_09
PCIAT.PCIAT_01
PCIAT.PCIAT_06
PCIAT.PCIAT_04
Physical.Weight
Physical.Height
Physical.BMI
PCIAT.PCIAT_07
Basic_Demos.Age
Physical.HeartRate
Physical.Systolic_BP
SDS.SDS_Total_T
SDS.SDS_Total_Raw
Physical.Diastolic_BP
CGAS.CGAS_Score
PCIAT.PCIAT_12
Variables
0
50
100
Mean Decrease in Gini



Linear Discriminant Analysis

Linear Discriminant Analysis (LDA) is a classification method that assumes the predictors for each class follow a multivariate Gaussian distribution with class-specific means and a shared covariance matrix. It estimates the posterior probabilities P(Y=k|X=x) using Bayes' theorem and assigns observations to the class with the highest posterior. LDA's decision boundaries are linear functions of the predictors, making it effective in separating classes under the Gaussian assumption. LDA operates by creating linear combinations of predictors that separates classes of a categorical response.

Prior to fitting the LDA model, we removed collinear variables from the data. We fit a model using 80 percent of the data as the training set and used the remaining 20 percent as the test set to predict what level `SII` might be classified as. We then created a confusion matrix for the training and testing sets.



When evaluating the LDA model on the training set, the model yielded a 1.19 percent error rate. When evaluated on the testing set, the model yielded a 0.72 percent error rate, which is evidence of strong predictive power.

Table 4: Train Confusion Matrix:

	0	1	2	3
0	1250	26	0	0
1	0	584	0	0
2	0	0	303	0
3	0	0	0	28

```
## [1] "Train Accuracy: 98.81 %"
```

```
## [1] "Train Error Rate: 1.19 %"
```

Table 5: Test Confusion Matrix:

	0	1	2	3
0	314	4	0	0
1	0	146	0	0
2	0	0	75	0
3	0	0	0	6

```
## [1] "Test Accuracy: 99.27 %"
```

```
## [1] "Test Error Rate: 0.73 %"
```

The LDA model performed better than expected, however, rhe minuscule error rates may be cause for speculation that there may be over-fitting present in the model. Even though collinearity was accounted for, there may be other variables that could cause this that escaped pruning. Failure to remove variables that were mostly NA values may be attribute to this as well. However, after re-evaluating the model with different techniques to handle NA values, the model behaved similarly, and the error rate was only slightly larger.

Conclusion

In conclusion, our exploration of various models to predict Severity Impairment Index (SII) and Sleep Disturbance Scale (SDS) scores highlights the challenges and strengths of different approaches. Random Forest and Linear Discriminant Analysis (LDA) showed strong predictive performance, with LDA achieving exceptionally low error rates, albeit with some concern for potential overfitting. Our data preprocessing, particularly handling missing values and addressing collinearity, played a crucial role in improving model reliability. Despite these efforts, differences in model performance suggest that further refinements, such as optimizing feature selection or exploring additional methods, could enhance predictions for these critical mental health indicators.